

Full Code & Project Audit

V4.1 — Sound & Motion release

A standalone HTML music-theory & harmony instrument for musicians and producers: an interactive Circle of Fifths, progression builder, weighted suggestion engine, production tab, instruments, i18n, motion design and — new in V4.1 — sound.

Date	June 2026
Version	V4.1 (live on GitHub Pages)
Stack	Vanilla JS + CSS + WebGL + Web Audio, no framework, no deps
Structure	Modular src/ tree compiled to one standalone HTML
Repository	peperonloo/Circle-of-Fifth-Git

1 — Executive read

Easy Fifth Circle has crossed from a promising single-file prototype into a real, modular product. The original audit's biggest worry — keeping a 3.9k-line HTML file stable while adding smart features — is structurally solved: the app is now a clean **src/** tree of 26 modules compiled by a repeatable build into one shareable standalone, auto-deployed to GitHub Pages.

The whole stabilization roadmap shipped (Mobile QA, Suggestion-Engine validation, CSS de-override, Overlay Manager, Test Suite), V4.0 added a layer of tasteful motion, and **V4.1 added sound** — you can finally hear what you build.

The next risk is no longer architecture — it is conceptual clarity and trust. One model issue (the Major/Minor wheel toggle overlapping the Mode selector) and a class of interaction bugs (invisible overlays, positional SVG selectors) show the interaction surface needs guard-rails the unit tests are only starting to cover.

2 — Snapshot (V4.1)

Metric	V3.17 (old)	V4.1 (now)
Structure	1 HTML file	src/ tree -> built standalone
Standalone size	231 KB	~216 KB
JavaScript	2,606 lines (1 file)	3,194 lines / 26 modules
CSS	1,076 lines	1,127 lines / 11 layers
Dev tests	~20	54, all green
Diagnostics	basic	runTests / mobileCheck / cssHealth / report
Duplicate ids	wheelPointer x2	0 (tested)
!important	n/a	153 (mostly defensive)
Deploy	manual file	GitHub Pages, build emits index.html
Audio	none	Web Audio FM Rhodes synth

Layers: DATA (constants, theory-data, i18n) · STATE (st, AppModel) · ACTIONS (ActionDispatcher) · ENGINES (theory / harmony / suggestion / audio) · RENDER (render-engine + renderers) · INTERACTIONS (wheel / builder / mobile) · OVERLAYS (OverlayManager) · TESTS (EFC_DEV).

3 — Strengths

- **Clear product identity:** a calm, visual harmony instrument — not a theory textbook or a DAW dashboard.
- **Modular and buildable:** logic / render / interaction are cleanly separated; one command produces a shareable standalone, and Pages auto-deploys from main.
- **The wheel is the anchor** and now feels alive (idle breathing, tonic heartbeat, selection ripple/bloom, colourful animated plasma).
- **Real musical intelligence:** the suggestion engine models functional harmony, cadences (ii-V-I), modal signatures, vamp/repetition and position, with short honest reasons and strength-by-size bubbles in degree order.
- **Sound (V4.1):** tap a chord to hear it, play the progression with a moving playhead — the single biggest trust upgrade.
- **Self-testing:** 54 dev checks plus mobile + CSS health, run on load and surfaced in a hidden dev panel — a strong regression base for a project this size.
- **Bilingual (EN/ES)** with strings centralised; **zero dependencies** keeps it tiny, portable and offline-friendly.

4 — Weaknesses & technical debt

- **Minor-view / Mode conflict (top issue):** two controls (the wheel Major/Minor toggle and the 7-mode dropdown) both govern major/minor-ness and aren't unified, and the model mixes relative vs parallel minor. This caused the wheel-jump bug (now patched) and still leaves the centre label / scale framing inconsistent. Needs one coherent model.
- **Interaction bugs slip past unit tests:** a closed overlay's invisible click-wall and a positional SVG selector that mis-filled the wheel were structural bugs 50+ unit tests didn't catch. Guard tests (H2) and id-based SVG selectors (H3) are now in but coverage is young.
- **!important density (153):** not competing layers anymore, but it makes the cascade harder to reason about. The bubble + popover layers are densest.
- **Animation budget:** several always-on animations + the WebGL plasma. Mitigated on mobile (30fps, pause-on-scroll, sheen off) but still needs a real low-end device FPS pass.
- **Production tab** is a parallel mode with the least test coverage; risks competing with the harmony core.
- **Suggestion engine** still leans on hand-tuned weights; genre and emotional context and longer look-ahead are not yet modelled.

5 — Deep dive: the minor / mode model

This is the most important finding, so it gets its own section. The wheel structure (outer ring = major keys, inner ring = relative minors) inherently encodes **relative** relationships. But two controls fight over major/minor-ness:

- The **Major/Minor toggle** on the wheel (a relative switch: C major <-> A minor).
- The **Mode dropdown** (parallel: keeps the tonic, changes the scale — C ionian -> C aeolian).

Because they were independent, the state could become self-contradictory: the wheel highlighted one sector while the centre label, the scale and the degrees described a different key. The recently-shipped **wheelKey()** fix makes the wheel rotate and highlight by the sector the selection actually occupies, so clicking no longer jumps — but the labelling/scale consistency still needs the model decision below.

Recommended resolution

Adopt one model and make every surface obey it. Recommended: treat the toggle as the relative major/minor switch (the circle-of-fifths-native reading), unify it with the Mode selector (Major = ionian, Minor = aeolian; the dropdown reflects and overrides), and drive the centre, scale, accidentals and degrees from a single source of truth so they can never disagree. Alternatively, commit to parallel minor and drop the inner-ring relabelling. Either is fine — the point is to pick one.

6 — Bugs found & fixed this cycle

Bug	Cause	Status
Wheel jumps / nothing selected in minor view	Rotation used the minor letter's major position, not the clicked sector	Fixed (wheelKey)
Invisible wall blocked the whole top bar on mobile	Closed guide cards kept pointer-events:auto at top:0	Fixed
Whitish fog over the wheel	A new circle shifted querySelectorAll('circle')[1]	Fixed (id-based)
Flat / static background	Shader mixed 3rd colour globally; palette was 3 oranges	Fixed
Wrong key signature in minor/modal	Computed from the wheel anchor's major signature	Fixed (parent-major)
Builder pills overlapped	Absolutely-positioned tools over the chord/degree	Fixed
Silent audio on mobile Chrome	AudioContext suspended; no gesture unlock	Fixed (global unlock)
Janky mobile scroll	Full-screen WebGL canvas repainting every frame	Mitigated

7 — Architecture verdict

Layer	State
DATA (constants, theory-data, i18n)	OK — no DOM access
STATE (st, AppModel)	OK — single source of truth
ACTIONS (ActionDispatcher)	OK — all interactions route through dispatch
ENGINES (theory / harmony / suggestion / audio)	OK — mostly pure logic
RENDER (render-engine + renderers)	Good — SVG selectors now id-based
INTERACTIONS (wheel / builder / mobile)	Watch — needs more hit-test / pointer tests
OVERLAYS (OverlayManager)	OK — one contract; should also own pointer-events
TESTS (EFC_DEV)	Strong unit base; interaction coverage young

8 — Complexity assessment

Current: medium-high for a standalone, zero-dependency app. It already combines SVG geometry, custom drag/roulette interaction, a state/action architecture, harmonic rules, WebGL background, Web Audio synthesis, motion design, i18n and responsive UI — a real product, not a widget.

Future: high if MIDI export, saved sessions, genre packs and richer audio land. This stays manageable because the foundations (modules, build, tests, deploy) are in place — the cost is now mostly in getting models (like minor/mode) conceptually right, not in wrangling one giant file.

Strategic call: the architecture is no longer the bottleneck. Invest the next few versions in conceptual clarity (the minor/mode model), depth (audio, genre) and interaction-test coverage — not in more structure.

9 — Roadmap

Near-term hardening (cheap, high trust)

- **H-now · Minor/Mode redesign:** one coherent model; centre, scale, accidentals and degrees from a single source.
- **Audio polish:** a Stop button, tempo control, velocity, and sound when tapping the on-screen piano/guitar.
- **Interaction-test coverage:** extend the guard tests (hit-testing, overlay contracts) to the builder and mode menu.
- **Low-end mobile FPS pass** and a global animation pause when the page is hidden.

Version	Goal	Why it matters
V4.2	Genre-aware suggestions	House / neo-soul / disco / ambient bias the scoring and the reasons
V4.3	Progression narrative V2	Turn a sequence into a story: beginning, lift, tension, release, home
V4.4	Export / Share	Copy progression, shareable URL (state in the hash), MIDI export, saved sessions
V4.5	Real-device mobile pass	iOS Safari / Android Chrome QA: drag vs scroll, theme wipe, FPS, audio
V5.0	Platform	PWA (installable, offline) + CI that gates every deploy on <code>EFC_DEV.runTests()</code>

10 — Ideas: what would make it incredible

- **Hear the harmony, not just see it** — richer voicings (7ths, inversions), a soft drum/bass groove from the Production tab, and a tempo so the progression plays in time.
- **Visual harmonic gravity** — subtle motion that shows why a chord wants to resolve somewhere (a refined version of the earlier arcs, kept tasteful).
- **Progressions become stories** — one short line of narrative per progression, driven by the existing R/T/M metrics.
- **Genre & emotion lenses** — the same key feels different through a house vs neo-soul vs cinematic lens.
- **Capture & share** — a link that reopens your exact wheel + progression; export to MIDI to continue in a DAW.
- **Always an instrument, never a textbook** — keep theory available on demand, never blocking the creative flow.

11 — Final recommendation

Two things next, in order: **(1) decide and implement one minor/mode model** so the wheel, centre, scale and degrees can never disagree; **(2) polish the new audio** (stop/tempo, playable instruments). The tool already looks and feels alive and now sounds — the remaining leap in trust is making the theory *coherent*

and the sound *expressive*. With that, this becomes a distinctive harmonic companion rather than just another circle-of-fifths app.

Easy Fifth Circle — V4.1 audit · generated for the project repository · live at peperonloo.github.io/Circle-of-Fifth-Git